Recitation 2: Proof Techniques

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Warmup

1. For every real number x, $x^2 \geq x$

   Solution: Let x be any fraction.

2. Prove that if n is an odd integer, then $3n + 7$ is an even integer.

   Solution: Let $n = 2k+1$ where $k \in \mathbb{Z}$. Then, $3n+7 = 3(2k+1)+7 = 6k+3+7 = 6k+10 = 2(3k+5)$, which must be even.

3. Prove that there is no smallest positive rational number.

   Solution: Suppose $r > 0$ is the smallest positive rational. Then $r = a/b$ where $a, b \in \mathbb{Z}$. Then, $a/2b$ must be rational too and also satisfies $0 < r/2 < r$, contradicting minimality.

Proof by Contraposition

1. Let $n \in \mathbb{Z}$. If $(3n + 2)^2$ is even, then n is even.

   Solution: We must show that if n is odd, then $(3n + 2)^2$ is odd. Then, $n = 2k + 1$ where $k \in \mathbb{Z}$ and $(3n+2)^2 = (3(2k+1)+2)^2 = (6k+5)^2 = 36k^2+60k+25 = 36k^2+60k+24+1 = 4(9k^2 + 15k + 6) + 1$ which is odd.

2. Prove that if ab is even (with $a, b \in \mathbb{Z}$), then a is even or b is even.

   Solution: We must show that if a and b are both odd, then ab is odd. With $a = 2k + 1$, $b = 2\ell + 1$, we have $(2k + 1)(2\ell + 1) = 4k\ell + 2k + 2\ell + 1 = 2(2k\ell + k + \ell) + 1$, which is odd.

3. If $n = ab$, where $a, b \in \mathbb{Z}$, then either $a \leq \sqrt{n+1}$ or $b \leq \sqrt{n+1}$.

   Solution: We must show that if $a > \sqrt{n+1}$ and $b > \sqrt{n+1}$, then $n \neq ab$. So, $ab = \sqrt{n+1} \times \sqrt{n+1} = n + 1 > n$, so $ab \neq n$.

Proof by Contradiction

1. Prove that if $r \in \mathbb{Q}$ and $s \notin \mathbb{Q}$, then $r + s \notin \mathbb{Q}$.

   Solution:. Suppose $r \in \mathbb{Q}$ and $s \notin \mathbb{Q}$. Let's assume that $r + s \in \mathbb{Q}$, meaning that there exist $a, b \in \mathbb{Z}$ such that $r+s = a/b$. Also, there exist $c, d \in \mathbb{Z}$ such that $r = c/d$ so $\frac{c}{d}+s = \frac{a}{b}$. Then,

   $s = \frac{a}{b} - \frac{c}{d} = \frac{ad-cb}{bd} \in \mathbb{Q}$, since $ad - cb, bd \in \mathbb{Z}$. Thus $s \in \mathbb{Q}$, which contradicts $s \notin \mathbb{Q}$.

2. Prove that there are no integers x and y such that $x^2 = 8y + 2$.

   Solution: Since $x^2 = 8y + 2 = 2(4y + 1)$, x^2 must be even so x must be even as well. Then, $x^2 = (2k)^2 = 4k^2$. If $x^2 = 4k^2 = 8y + 2 \rightarrow 2k^2 = 4y + 1$, which implies that $2k^2$ is odd.

Proof by Induction

1. Prove that for all $n \geq 1$, $n! \geq 2^{n-1}$.

   Solution: Base case $n = 1$: $1! = 1 \geq 2^0$. Next, we assume that the claim holds for some $k \geq 1$ (i.e., $k! \geq 2^{k-1}$) and show that $(k + 1)! \geq 2^{k-1+1} = 2^k$.

   $(k + 1)! = (k + 1)k!$ and $(k + 1)k! \geq (k + 1)2^{k-1}$. Since $k \geq 1$, $k + 1 \geq 2$. Therefore,

   $(k + 1)2^{k-1} \geq 2 \cdot 2^{k-1} = 2^k$, as desired.

2. Prove that for all $n \geq 1$, $10^n - 1$ is divisible by 9, i.e., show that $9 \mid (10^n - 1)$.

   Solution For the base case of $n = 1$ we have $10^n - 1 = 9$, which is clearly divisible by 9.

   Suppose the claim holds for some $n = k$. We will show that it holds for $k + 1$. Since it holds for k, we have that $10^k - 1 = 9\ell$ for some $\ell \in \mathbb{Z}$. Therefore,

   $10^{k+1} - 1 = 10 \cdot 10^k - 1 = 10(9\ell + 1) - 1 = 9(10\ell + 1)$, which is divisible by 9.

Proof by Parts

1. Prove that there is no integer x such that $x^2 \equiv 2 \pmod 3$.

   Solution: Any integer x is either $0, 1$, or 2 mod 3. Let's consider each case individually.

   (1) If $x \equiv 0 \pmod 3$ then $x = 3k$ for some $k \in \mathbb{Z}$ so $x^2 = 0 \pmod 3$.
   (2) If $x \equiv 1 \pmod 3$ then $x = 3k + 1$ for some k, so $x^2 = 9k^2 + 6k + 1 \equiv 1 \pmod 3$.
   (3) If $x \equiv 2 \pmod 3$ then $x = 3k+2$ for some k so $x^2 = 9k^2+12k+4 = 3(3k^2+4k+1)+1 \equiv 1 \pmod 3$.

   This covers every part of $x \pmod 3$ and in none of these cases do we have $x^2 \equiv 2 \pmod 3$.

2. Let $a, b \in \mathbb{Z}$. If $a^2 + b^2$ is odd, then $a + b$ is also odd.

   Solution: Let's suppose that $a+b$ is even. By the contrapositive, we must show that a^2+b^2 is also even. If $a+b$ is even, both a and b must have the same parity since $a+b = 2k+2\ell = 2(k+\ell)$ or $a + b = 2k + 1 + 2\ell + 1 = 2(k + \ell + 1)$ where $k, l \in \mathbb{Z}$. Consider each part individually.

   (1) a and b are both even: $a = 2k$ and $b = 2\ell$ so $a^2+b^2 = (2k)^2+(2\ell)^2 = 4k^2+4\ell^2 = 4(k^2+\ell^2)$ which is even.
   (2) a and b are both odd: $a = 2k + 1$ and $b = 2\ell + 1$ so $a^2 + b^2 = (2k + 1)^2 + (2\ell + 1)^2 = 4k^2 + 4k + 1 + 4\ell^2 + 4\ell + 1 = 2(2k^2 + 2k + 2\ell^2 + 2\ell + 1)$ which is even.

   Since $a^2 + b^2$ is even when $a + b$ is even in both cases, the contrapositive is also true.